Linear Regression

Recap on Logistic Regression / Optimization

Binary classification with $\mathscr{D} = \{x_i, y_i\}_{i=1}^n, x_i \in \mathbb{R}^d, y_i \in \{-1,1\}$

Logistic Regression asumes $P(y \mid x; w) = \dfrac{1}{1 + \exp(-y(w^\top x))}$

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Given a test example x_{test}, we can make prediction:

$$\hat{y} = \begin{cases} +1 & P(+1 \mid x_{test}; \hat{w}) > 0.5 \\ -1 & \text{else} \end{cases}$$

Recap on Logistic Regression / Optimization

$$\arg\min_{w} \sum_{i=1}^{n} \ln(1 + \exp(-y_i(w^\top x_i)))$$

Logistic Regression with SGD as the optimizer:

Initialize $w^0 = \mathbf{0}$

While not converged:

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Compare this to Perceptron!

Outline for Today

1. Intro on Linear Regression

2. Normal equation for linear Regression

3. Interpretation of Linear Regression using MLE / MAP

Ex: Predicting the house price

Dataset:

Living area (feet^2)	Price (1000\$s)
2104	400
1600	330
2400	369
1416	232
3000	540
$\vdots$	$\vdots$
x	y

(Example from Stanford CS229)

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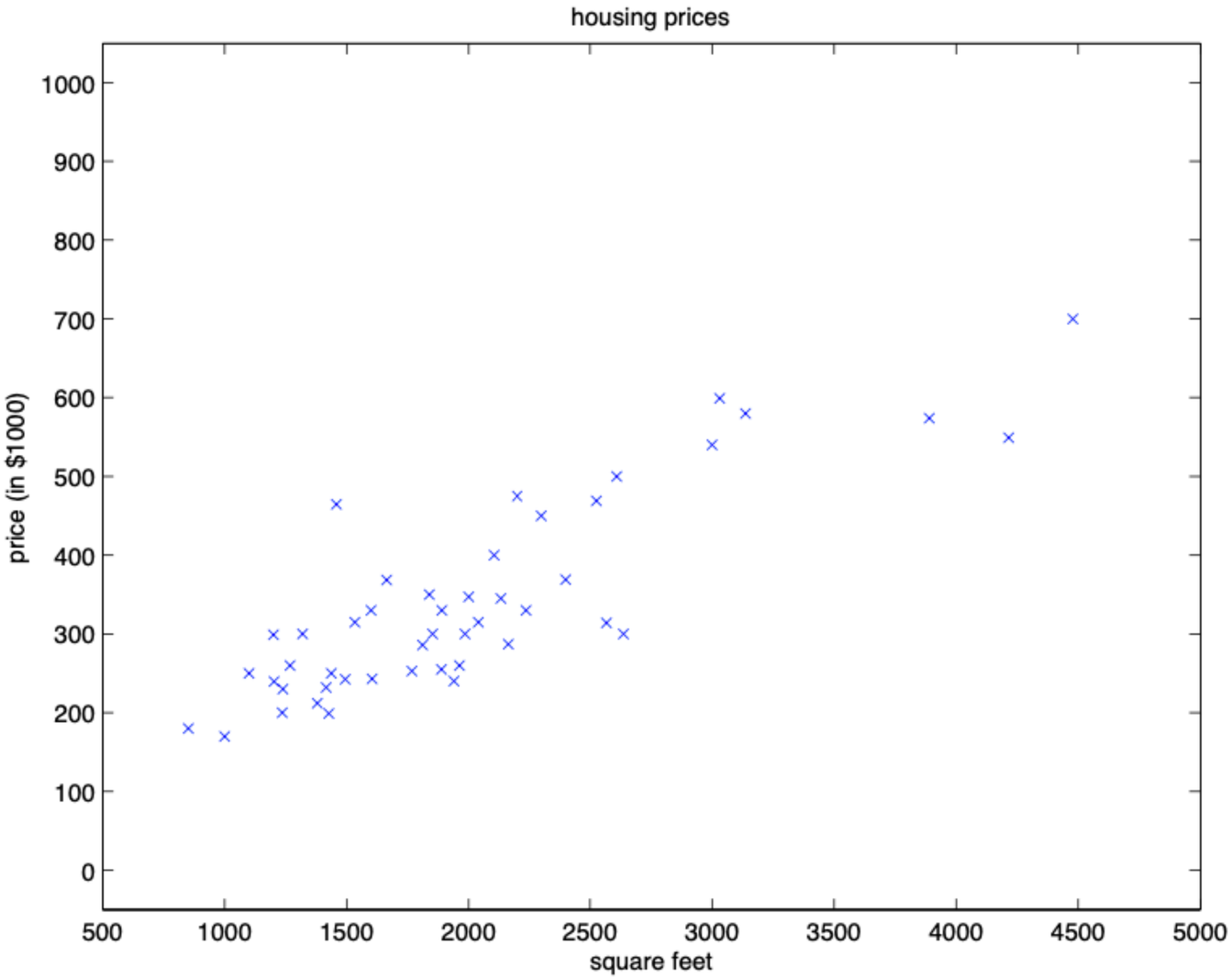

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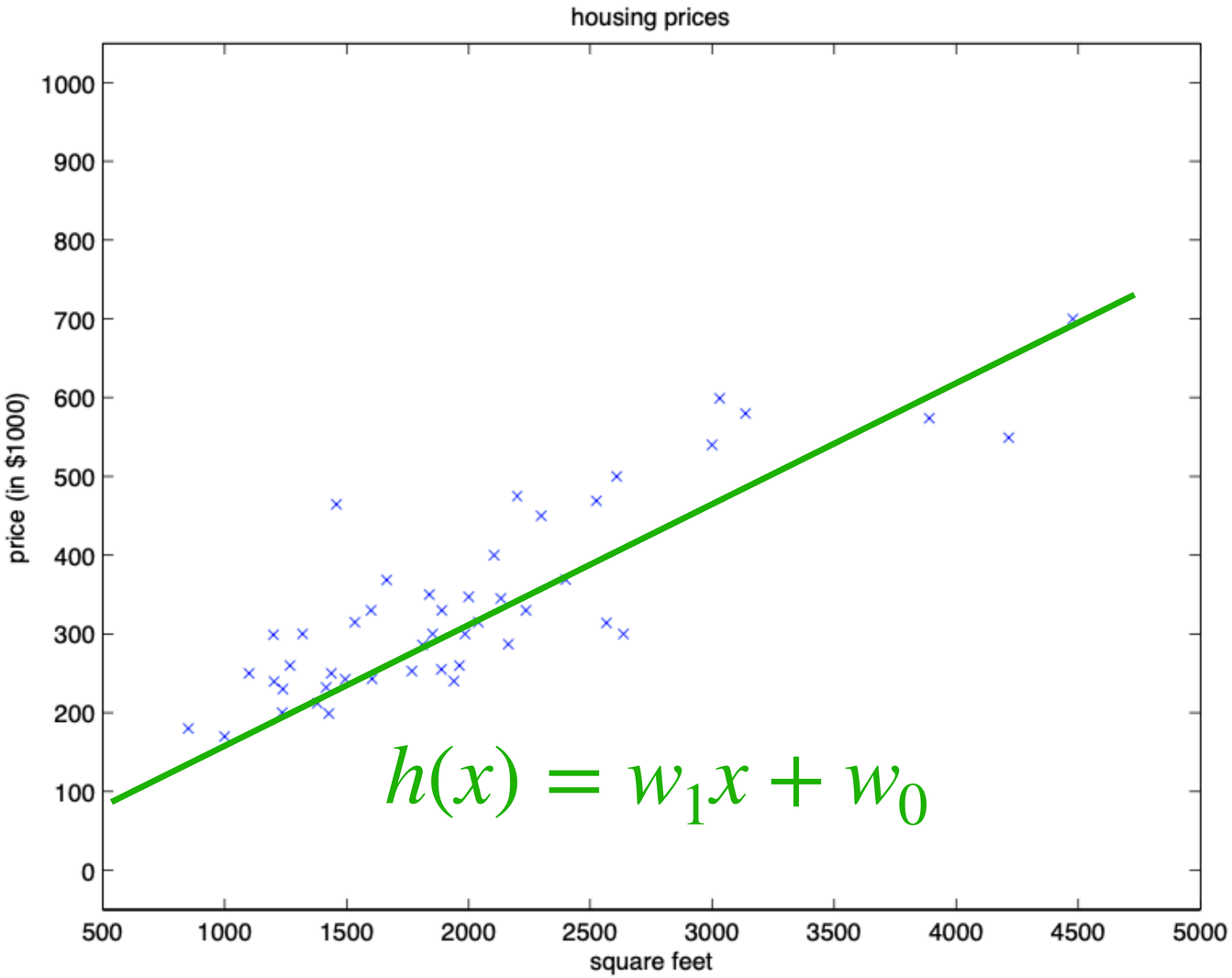

(Example from Stanford CS229)

Ex: Predicting the house price (2d case)

Dataset:

Living area (feet^2)	#bedrooms	Price (1000$s)
2104	3	400
1600	3	330
2400	3	369
1416	2	232
3000	4	540
$\vdots$	$\vdots$	$\vdots$
$x[1]$	$x[2]$	y

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Goal: finding the linear function

$$h(x) = w_1 x[1] + w_2 x[2] + w_0$$

that fits the data well

(Example from Stanford CS229)

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As usual, we append 1 to the feature, i.e.,

$$x = \begin{bmatrix} x[1] \\ x[2] \\ 1 \end{bmatrix}$$

So the linear function can be written as:

$$h(x) = w^\top x$$

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1. Intro on Linear Regression

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Mathematical formulation of linear regression

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Q: can we use absolute loss, i.e., $|w^\top x - y|$?

Q: can we use $(w^\top x - y)^3$?

Mathematical formulation of linear regression

Formulating the optimization problem:

$$\arg\min_{w} \sum_{i=1}^{n} (w^\top x_i - y_i)^2$$

Linear regression solution

$$\arg\min_{w} \sum_{i=1}^{n} (w^\top x_i - y_i)^2$$

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$$\Rightarrow \arg\min_{w} \|X^\top w - Y\|_2^2$$

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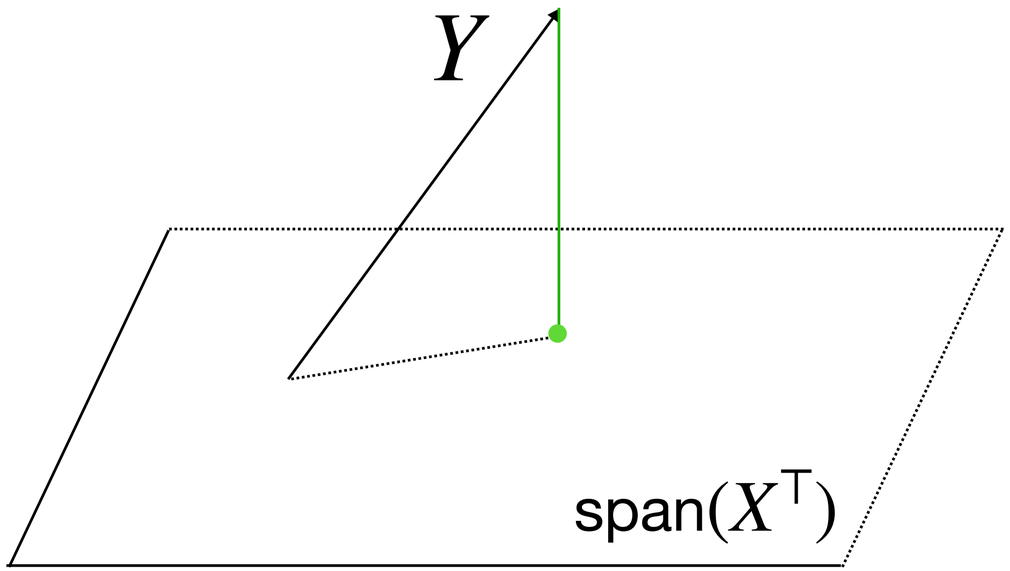

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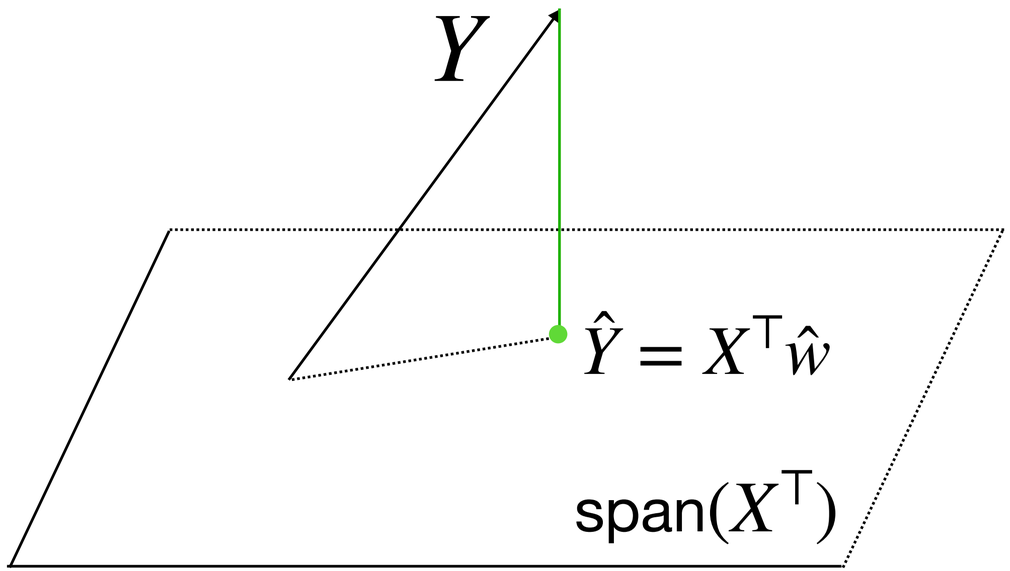

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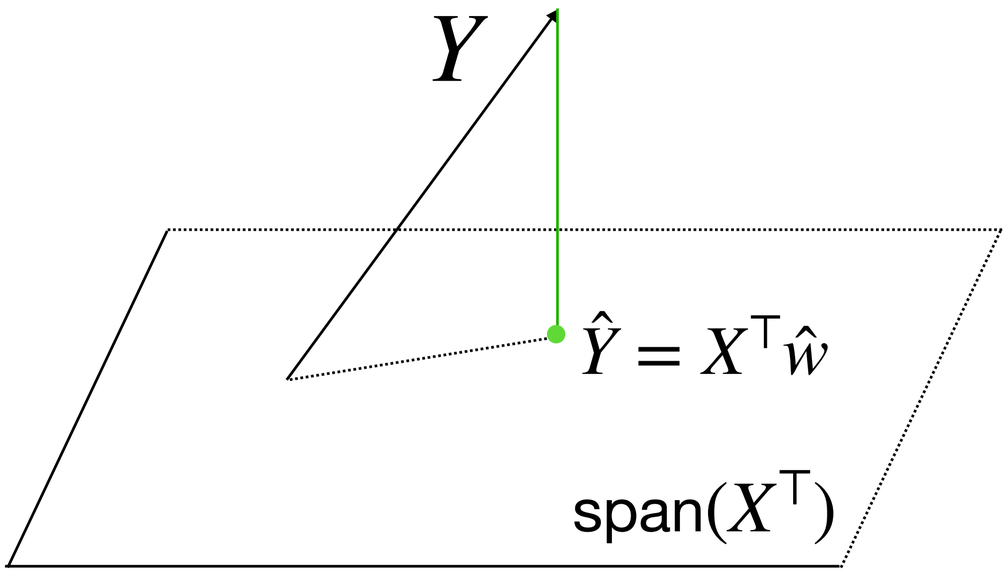

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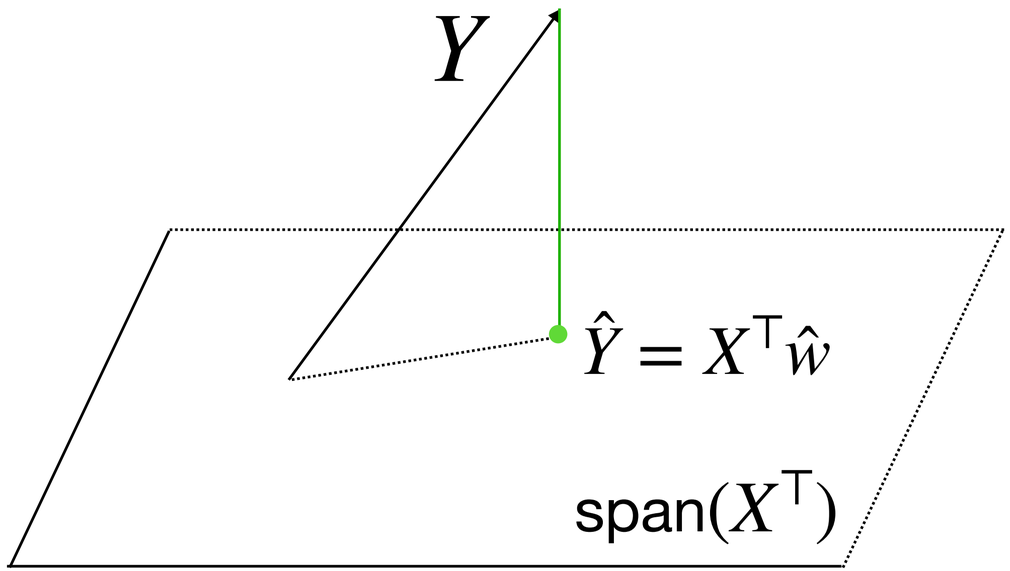

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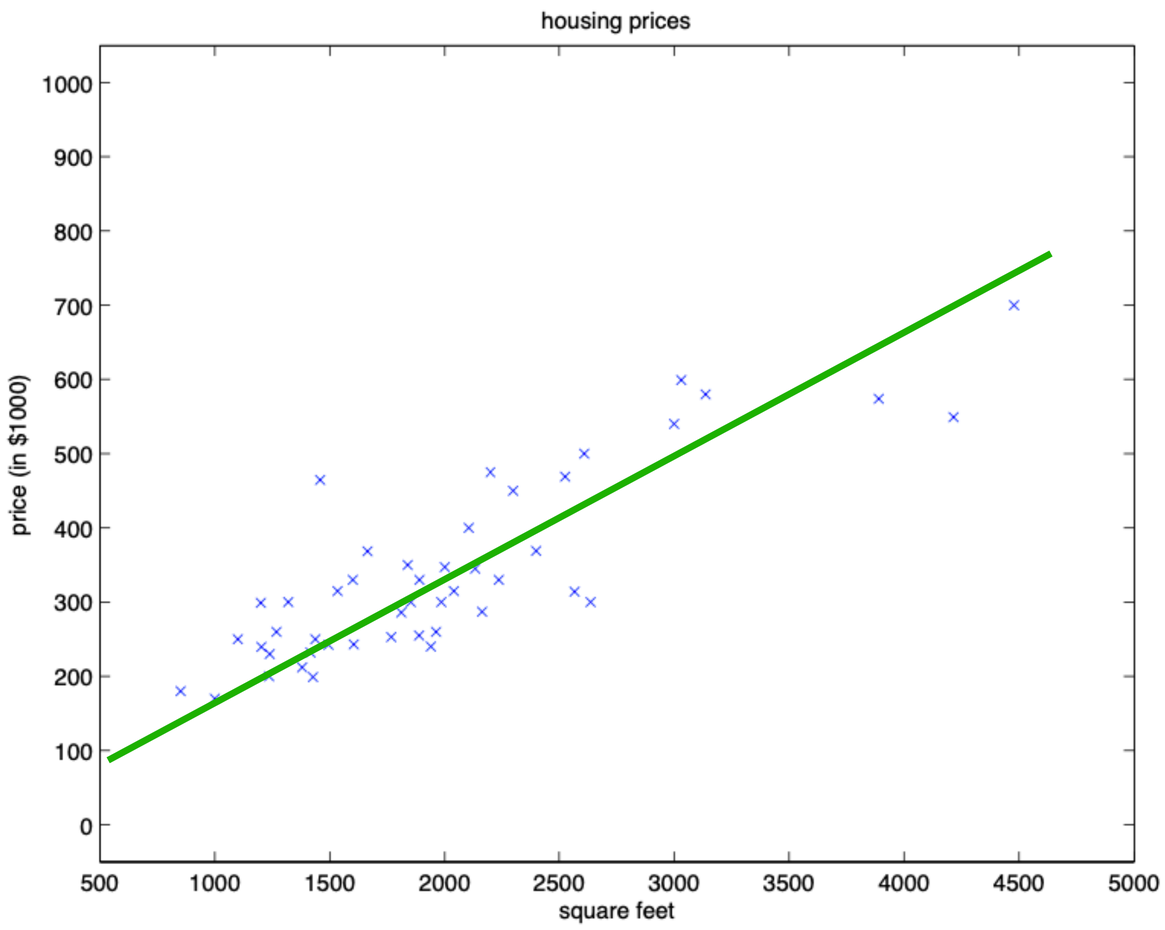
(We will talk about regularization soon)

Prediction using linear regression

Once we learned $\hat{w}$, we can use it to make prediction on any new feature x



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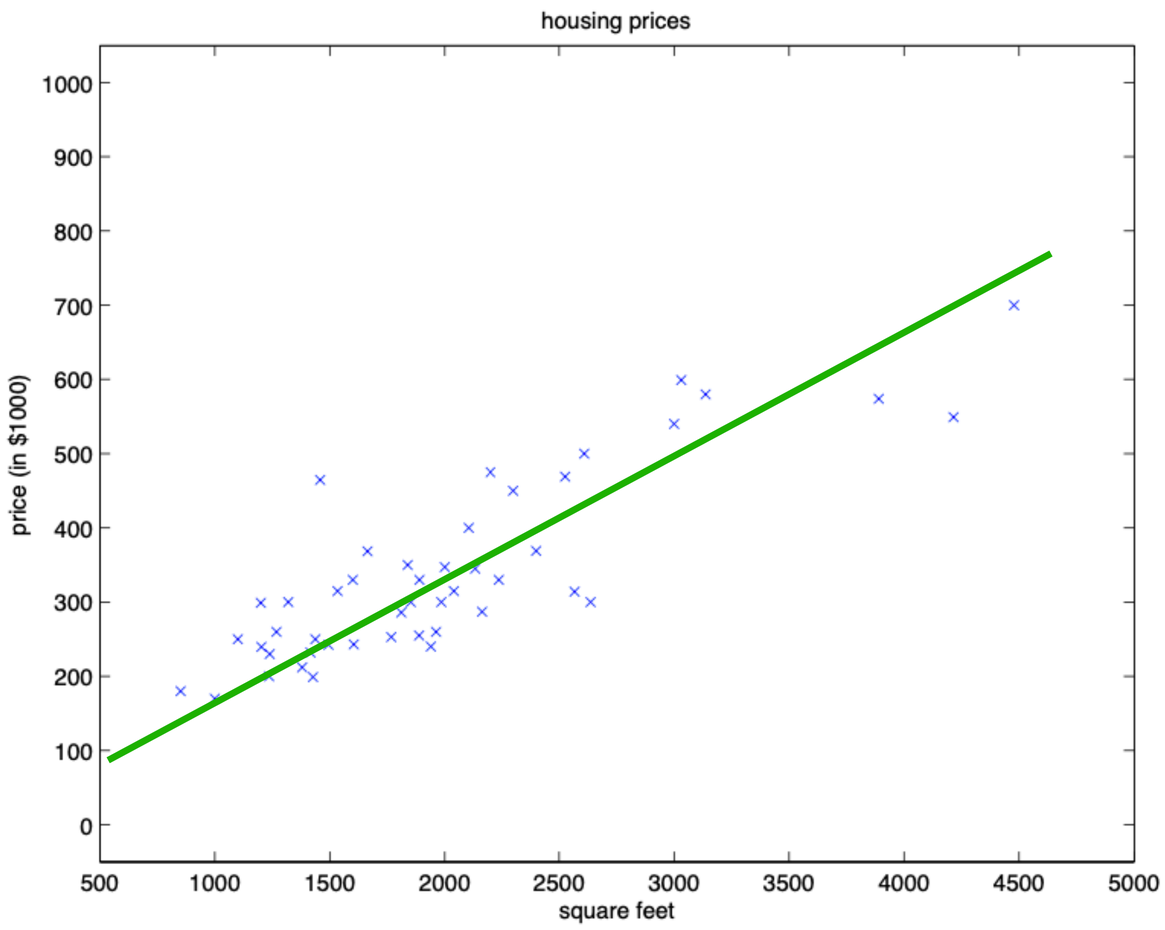

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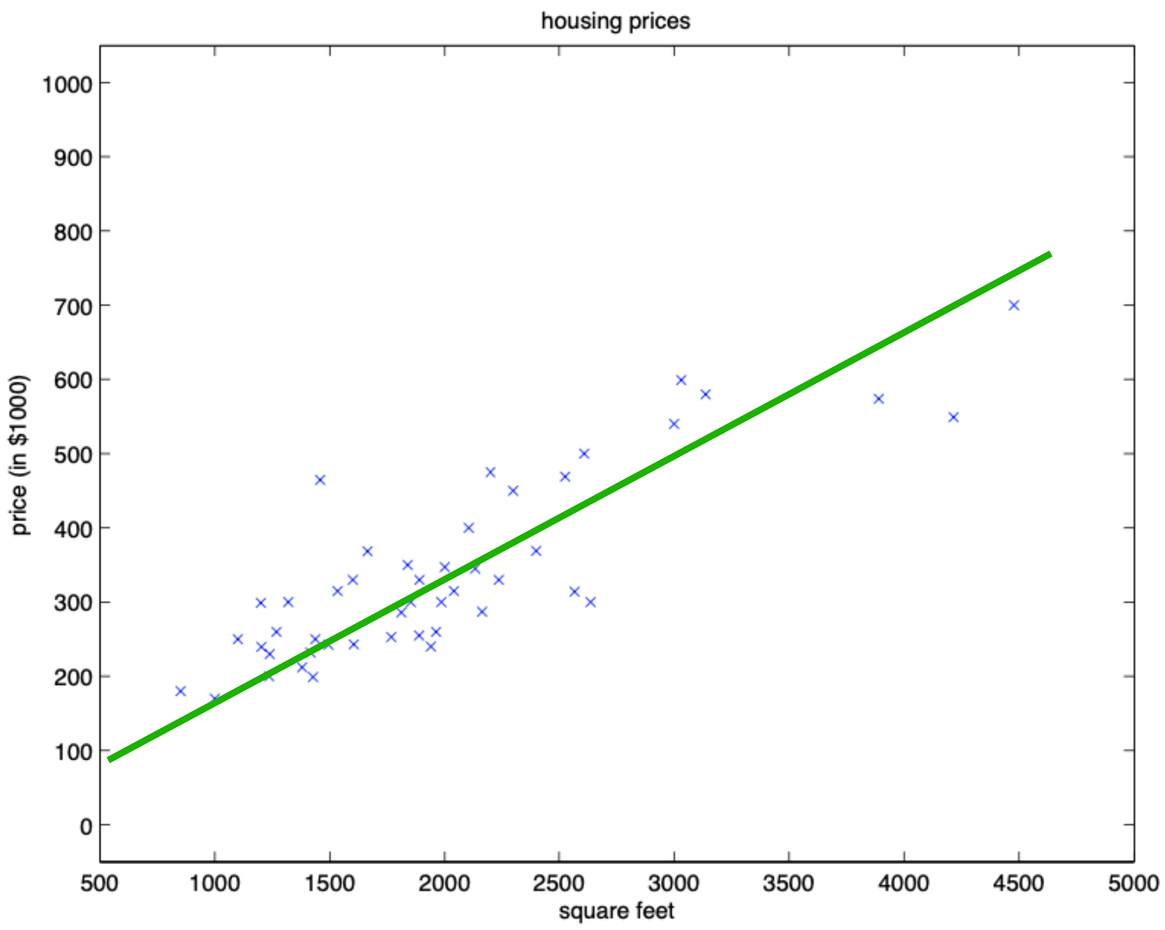

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Derive Linear regression via Maximum Likelihood Estimation

Assume $P(y \mid x; w) = \frac{1}{Z} \exp\left(-\frac{1}{2}(y - x^\top w)^2/\sigma^2\right)$, i.e., $y = w^\top x + \epsilon$, $\epsilon \sim \mathcal{N}(0,\sigma^2)$

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Derive Linear regression via MAP

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To use MAP, we need to define a prior over w, we use Gaussian as well here:

$$w \sim \mathcal{N}(0,r^2 I)$$

Derive Linear regression via MAP

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$$= \arg\min_w \frac{\sigma^2}{r^2} w^\top w + \sum_{i=1}^{n} (w^\top x_i - y_i)^2 \quad = \arg\min_w \lambda \|w\|_2^2 + \sum_{i=1}^{n} (w^\top x_i - y_i)^2$$

Ridge Linear Regression

$$\arg\min_{w} \lambda \|w\|_2^2 + \sum_{i=1}^{n} (w^\top x_i - y_i)^2$$

In this case, we can derive a closed-form solution as well:

$$\hat{w} = (XX^\top + \lambda I)^{-1} XY$$

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Note that it works even $XX^\top$ is not full rank

Summary for today

1. Linear regression, Normal equation, and MLE / MAP interpretation